FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 — SUB-STRAND 3.2: CURRENT ELECTRICITY	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION TASK — "The Flickering Lights of Huruma Estate"

Your goal in this Final Explanation is to answer the driving question using everything you have learned across all 12 lessons in this unit. Think back to the anchoring phenomenon: when power was restored after a Kenya Power blackout in Huruma Estate, the lights glowed dimly, the kettle heated slowly, and a bulb on a long thin extension cord was always dimmer than normal. You now have the scientific tools to fully explain WHY this happened — and to go further and design a solution.

Write your Final Explanation in full sentences and paragraphs. Use correct scientific vocabulary throughout. Support every claim you make with scientific reasoning and evidence from your investigations. Where asked, include a clearly labelled diagram or circuit model.

Your response must address ALL FIVE sections below. Each section builds on the previous one — together, they form your complete answer to the driving question:

"Why do electrical devices in our homes sometimes receive less energy than expected, and how can we design circuits to deliver the right amount of electrical energy safely and efficiently?"

Read each section prompt carefully before you begin. Use the exemplar responses as a guide to the level of detail and reasoning expected.

Section 1 — Voltage, Current, and Resistance: Building Your Model	
Explain the relationship between voltage (V), current (I), and resistance (R) in an electric circuit. In your answer: – **State Ohm's Law in words and as an equation.** – **Define each quantity in**	Ohm's Law states that the current flowing through a conductor is directly proportional to the voltage across it, provided temperature and other physical conditions remain constant. We write this as $V = IR$, where V is voltage measured in volts (V), I is current measured in amperes (A), and R is resistance measured in ohms (Ω). Voltage is the electrical 'push' that drives charges around a circuit — think of it like the pressure that pushes water through a pipe. Current is the rate at which charge flows — the actual movement of electrons through the wire. Resistance is anything that opposes that flow, like a narrow section of pipe slowing down water.

your own words, including its SI unit. – **Use the Huruma Estate phenomenon as your context: explain what was happening to the voltage, current, and resistance in the circuit when the lights first came back on dimly after the blackout.** – **Use at least ONE numerical example with a calculation to support your explanation.**	When power was first restored in Huruma Estate, the supply voltage from Kenya Power was lower than the normal 240 V — this happens during the transition period as the grid re-stabilises. If we say the voltage dropped to 120 V, and the resistance of a bulb's filament is approximately 960 Ω at operating temperature, then by Ohm's Law: I = V ÷ R = 120 ÷ 960 = 0.125 A. At normal voltage, I = 240 ÷ 960 = 0.25 A. With only half the normal current flowing, the bulb received far less energy per second — which is exactly why it glowed dimly. The voltage, current, and resistance are always linked: if voltage drops and resistance stays roughly the same, current drops too, and the device behaves differently than expected.

Section 2 — Resistance in Circuits: Series, Parallel, and the Extension Cord Problem

Explain how resistance behaves differently in series and parallel circuits. Then use this knowledge to explain why the neighbour's bulb connected via a long, thin extension cord was ALWAYS dimmer — even after the mains voltage returned to normal 240 V. – **State and apply the rules for total resistance in series and parallel circuits.** – **Explain the concept of internal resistance and how it relates to the Huruma scenario.** – **Include a clearly labelled circuit diagram (drawn or described in detail) showing the extension cord as a resistor in series with the bulb.** – **Use a calculation to show how the extension cord reduces the voltage available to the bulb.**	In a series circuit, resistors are connected end-to-end so the same current flows through each one. The total resistance is simply the sum of all individual resistances: $R_{total} = R_1 + R_2 + R_3 + \ldots$ This means adding more resistance in series always increases the total resistance and reduces the current. In a parallel circuit, resistors are connected across the same two points so they share the voltage. The total resistance is found using $1/R_{total} = 1/R_1 + 1/R_2 + \ldots$, and total resistance is always LESS than the smallest individual resistor. This is why parallel circuits are used in home wiring — each appliance gets the full mains voltage. The neighbour's extension cord acted as an extra resistor placed IN SERIES with the bulb. Long, thin wires have higher resistance than short, thick ones, because resistance depends on length (longer = more resistance) and cross-sectional area (thinner = more resistance). Suppose the bulb has resistance 960 Ω and the extension cord has resistance 40 Ω. The total series resistance is 960 + 40 = 1000 Ω. Using V = IR with the mains at 240 V: I = 240 ÷ 1000 = 0.24 A. The voltage dropped across the extension cord is: V_{cord} = 0.24 × 40 = 9.6 V. So the voltage actually reaching the bulb is only 240 − 9.6 = 230.4 V — a loss of 9.6 V. This 'voltage drop' across the cord means the bulb receives less power and appears dimmer. This is also related to internal resistance — the source of electricity (Kenya Power's supply, or a battery) has its own small internal resistance. When the grid was first restored, internal resistance in the supply and the distribution cables was effectively higher, causing a larger voltage drop before electricity even reached the homes of Huruma Estate. As the system stabilised, the effective internal resistance fell and full voltage was delivered.

Section 3 — Electrical Power and Energy: Why the Kettle Was Slow	
Explain the concepts of electrical power and electrical energy, and use them to explain why the kettle in Huruma Estate took longer than usual to boil water when the lights were dim. **– State the equations for electrical power ($P = IV$, $P = V^2/R$, $P = I^2R$) and electrical energy ($E = Pt$).** **– Explain in your own words what each equation tells us about how electrical devices consume energy.** **– Use a calculation to compare the power delivered to the kettle at reduced voltage (e.g. 120 V) versus normal voltage (240 V), and estimate how much longer it would take to boil the same amount of water.** **– Connect your answer back to the Huruma phenomenon clearly.**	Electrical power is the rate at which electrical energy is converted into other forms (heat, light, sound, etc.). It is measured in watts (W), where 1 W = 1 joule per second. The key equations are: – $P = IV$ (power equals current times voltage) – $P = V^2/R$ (useful when voltage and resistance are known) – $P = I^2R$ (useful when current and resistance are known) Electrical energy is the total amount of energy transferred over a period of time: $E = Pt$, measured in joules (J) or, for household billing purposes, kilowatt-hours (kWh). Consider the kettle in Huruma Estate. A typical Kenyan electric kettle is rated at 2400 W when connected to 240 V. Its resistance is: $R = V^2/P = (240)^2 \div 2400 = 57600 \div 2400 = 24\ \Omega$. When the voltage dropped to 120 V after the blackout: $P = V^2/R = (120)^2 \div 24 = 14400 \div 24 = 600$ W. The kettle was operating at only 600 W instead of 2400 W — just one quarter of its normal power! Since energy needed to boil the water (E) stays the same, and $E = Pt$, the time needed is $t = E \div P$. At normal power: if it normally takes 3 minutes (180 s), the energy needed is $E = 2400 \times 180 = 432{,}000$ J. At reduced power: $t = 432{,}000 \div 600 = 720$ seconds = 12 minutes. The kettle would take FOUR TIMES as long to boil the same water. This perfectly explains the Huruma residents' experience — reduced voltage meant drastically reduced power, which meant the kettle seemed to 'barely heat' the water.

Section 4 — Safe and Efficient Circuit Design: Solving the Huruma Problem	
Using your knowledge of circuits, resistance, power, and energy, propose and justify a circuit design solution that would help residents of Huruma Estate receive electrical energy safely and efficiently — even when supply voltage fluctuates or extension cords are needed. **– Explain why parallel circuits are preferred over series circuits for home wiring.** **– Discuss the role of fuses, circuit breakers, and**	For home wiring, parallel circuits are far superior to series circuits for one key reason: in a parallel circuit, every appliance is connected directly across the full mains voltage (240 V in Kenya). This means each device operates at its designed voltage regardless of what other devices are connected. In a series circuit, the voltage is shared between all appliances — switching on an extra device would reduce the voltage (and brightness/power) of all other devices. This is exactly what the extension cord was causing in Huruma: it introduced unwanted series resistance, reducing the voltage reaching the bulb. Safety in home circuits depends on three critical components: 1. Fuses — a thin wire that melts and breaks the circuit if current exceeds a safe level (e.g. a 13 A fuse protects a circuit from dangerous overloads). Once melted, it must be replaced. 2. Circuit breakers — automatic switches that trip (open) when current is too high. Unlike fuses, they can be reset after the fault is fixed. Modern Kenyan homes use MCBs (Miniature Circuit Breakers). 3. Earthing — connects the metal casing of appliances to the ground. If a live wire accidentally touches the casing, current flows safely to earth instead of through a person, preventing electrocution.

earthing in electrical safety. – **Recommend at least TWO specific, practical improvements a Huruma Estate resident could make (e.g. regarding wire gauge/thickness, circuit configuration, or use of a voltage stabiliser).** – **Draw or describe a labelled circuit diagram of a safe home circuit for a single room, including a switch, fuse, two appliances in parallel, and an earth connection.**	Practical recommendations for Huruma Estate residents: – Use thick-gauge (low-resistance) extension cords rated for the appliances being used. A thicker wire has lower resistance, causing less voltage drop and less wasted energy as heat in the cord. – Install a voltage stabiliser (automatic voltage regulator, AVR) — this device monitors incoming voltage and corrects fluctuations, ensuring appliances always receive close to 240 V even during grid instability after load-shedding. A safe single-room circuit would show: the live wire from the mains passing through a fuse, then through a switch, connecting to two appliances (e.g. a bulb and a fan) wired in PARALLEL across the live and neutral wires. Each appliance has its metal casing connected by a green/yellow earth wire to the earth pin. Both appliances operate at 240 V, and the fuse protects the entire circuit from overcurrent.

Section 5 — Answering the Driving Question: Your Full Scientific Explanation	
Now bring everything together. Write a coherent, well-structured scientific explanation (minimum 3 paragraphs) that directly and fully answers the driving question: **'Why do electrical devices in our homes sometimes receive less energy than expected, and how can we design circuits to deliver the right amount of electrical energy safely and efficiently?'** **Your answer must:** – **Reference the Huruma Estate phenomenon specifically (the dim lights, the slow kettle, the dimmer bulb on the extension cord).** – **Use all key vocabulary from the unit: voltage,**	Electrical devices in our homes sometimes receive less energy than expected because of resistance — either in the supply lines, in extension cords, or within the power source itself. Resistance opposes the flow of current, and by Ohm's Law ($V = IR$), any resistance in the circuit causes a voltage drop — meaning less voltage actually reaches the device. In Huruma Estate, when Kenya Power restored electricity after the blackout, the supply voltage was initially below the normal 240 V as the grid re-stabilised. This low voltage meant that, by $P = V^2/R$, both the bulb and the kettle received far less power than they were designed for. The bulb glowed dimly because less electrical energy was being converted to light and heat per second. The kettle took much longer to boil because at lower power, it transferred energy to the water at a much slower rate — our calculation showed it could take up to four times longer. The neighbour's bulb on the long thin extension cord was ALWAYS dim — even after full voltage was restored — because the cord itself added resistance in series with the bulb. This created a voltage divider effect: some of the 240 V was 'used up' pushing current through the cord's resistance, leaving less voltage for the bulb. The thinner and longer the cord, the greater its resistance, and the greater the voltage drop across it. This is the same principle as internal resistance in a battery or power supply: any resistance between the source and the load steals some of the available voltage before it can do useful work. To design circuits that deliver the right amount of electrical energy safely and efficiently, we use parallel wiring (so every appliance gets the full supply voltage independently), low-resistance wiring (thick-gauge cables for high-power appliances), fuses and circuit breakers (to interrupt dangerously high currents before they damage devices or start fires), and earthing (to provide a safe path for fault currents and protect people from electrocution). A Huruma Estate resident could improve their situation immediately by replacing thin extension cords with properly rated thick-gauge cables and by fitting a voltage stabiliser to protect sensitive appliances during grid fluctuations. At the start of this unit, I thought electricity either worked or it didn't — that a device was either on or off, receiving power or not. I now understand that electrical energy delivery exists on a spectrum controlled by voltage, current, and resistance. The same source can deliver vastly different amounts of power to a device depending on the resistance of the circuit it travels through. Every

current, resistance, Ohm's Law, series circuit, parallel circuit, internal resistance, power, energy, voltage drop, fuse, circuit breaker, earthing. – **Show how the science you have learned connects across all four previous sections.** – **End with a statement of what you now understand that you did NOT understand at the start of this unit.**	wire, every connection, and every component matters — and good circuit design is what ensures that the energy generated at a power station like Olkaria Geothermal Plant in the Rift Valley actually reaches a kettle in Huruma Estate efficiently, safely, and at the right level.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy & Use of Ohm's Law	Correctly states and applies Ohm's Law ($V = IR$) and power equations ($P = IV$, $P = V^2/R$, $P = I^2R$) in all contexts. All calculations are accurate and clearly shown with correct SI units. Explains voltage, current, and resistance with precision and no misconceptions.	Correctly states Ohm's Law and power equations and applies them in most contexts. Calculations are mostly accurate with minor arithmetic errors. Definitions are correct but may lack depth in one area.	States Ohm's Law but applies it inconsistently or with errors. Calculations contain significant mistakes or are missing units. Definitions of voltage, current, or resistance are incomplete or contain misconceptions.
Explanation of Series vs. Parallel Circuits & Voltage Drop	Clearly and accurately explains series and parallel circuit rules. Correctly identifies the extension cord as a series resistor causing voltage drop, supported by accurate calculation. Connects internal resistance concept to the Huruma phenomenon with precision. Circuit diagram (described or drawn) is complete and correctly labelled.	Explains series and parallel circuits correctly. Identifies the extension cord as the cause of the voltage drop and provides a mostly correct calculation. Internal resistance is mentioned but explanation may lack full depth. Circuit diagram is mostly correct with minor omissions.	Shows partial understanding of series and parallel circuits. The role of the extension cord is identified but the explanation of voltage drop is unclear or unsupported by calculation. Internal resistance is not addressed or is confused with another concept. Circuit diagram is missing or has significant errors.
Application of Electrical Power & Energy to the Phenomenon	Accurately explains power and energy using all three power equations. Uses a correct, clearly worked calculation to show how reduced voltage dramatically reduces power (e.g. to one-quarter) and multiplies the time needed to boil water. Directly and explicitly connects this to the Huruma kettle and dim lights	Explains power and energy correctly and uses at least one power equation accurately. Calculation shows reduced power at lower voltage with mostly correct reasoning. Connection to the Huruma phenomenon is present but could be more explicit or detailed.	Attempts to explain power or energy but equations are misused or calculations are incorrect. The link between reduced voltage and longer boiling time is stated but not supported with evidence or calculation. The Huruma phenomenon is mentioned but not meaningfully explained.

	phenomenon.		
Safe and Efficient Circuit Design	Provides a well-reasoned justification for parallel home wiring. Accurately explains the function of fuses, circuit breakers, AND earthing. Recommends at least two specific, practical and scientifically justified improvements for Huruma residents. Described/drawn circuit diagram includes all required components (switch, fuse, two parallel appliances, earth connection) correctly.	Justifies parallel wiring for home use. Explains at least two of three safety components (fuse, circuit breaker, earthing) accurately. Offers at least one practical recommendation with reasonable justification. Circuit diagram includes most required components with minor errors or omissions.	Mentions parallel circuits or safety devices but explanations are vague or partially incorrect. Fewer than two safety components are explained, or explanations contain errors. Recommendations are generic or not scientifically justified. Circuit diagram is missing key components or is incorrectly labelled.
Coherence of Final Explanation & Use of Scientific Vocabulary	The final section provides a fluent, well-structured, multi-paragraph response that explicitly and fully answers the driving question. All key unit vocabulary (voltage, current, resistance, Ohm's Law, series, parallel, internal resistance, power, energy, voltage drop, fuse, circuit breaker, earthing) is used correctly and in context. The Huruma phenomenon is woven throughout. A meaningful reflection on conceptual change is included.	The final section addresses the driving question clearly and uses most key vocabulary correctly. The Huruma phenomenon is referenced but may not be consistently connected across all points. Reflection on prior understanding is present but may be brief or general.	The final section attempts to answer the driving question but the response is incomplete, disjointed, or relies on everyday language rather than scientific vocabulary. Key terms are missing, misused, or used without explanation. The Huruma phenomenon is mentioned only superficially. Reflection on conceptual change is absent or trivial.